



JPK

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KEMENTERIAN SUMBER MANUSIA
ARAS 7 & 8 BLOK D4, KOMPLEKS D
PUSAT PENTADBIRAN KERAJAAN
PERSEKUTUAN
62530 PUTRAJAYA**

**KERTAS PENERANGAN
(INFORMATION SHEET)**

KOD DAN NAMA PROGRAM / <i>PROGRAM'S CODE AND NAME</i>	Z-009-2: 2015CORE ABILITIES	
TAHAP / LEVEL	2 (TWO)	
NO. DAN TAJUK KEBOLEHAN TERAS/CORE ABILITY NO. AND TITLE	CA04 HEALTH, SAFETY AND ENVIRONMENTAL ADAPTATION	
NO. DAN PENYATAAN KEBOLEHAN / ABILITY NO. AND STATEMENT	04.01 DEMONSTRATE HEALTH COMPLIANCE 04.02 DEMONSTRATE SAFETY COMPLIANCE 04.03 DEMONSTRATE ENVIRONMENTAL COMPLIANCE	
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TAJUK/TITLE : SAFETY AND HEALTH WORKPLACE APPLICATION

TUJUAN/PURPOSE :

This Information Sheet is intended to describe on the ability to demonstrate level of understanding in Health, Safety and Environment adaptation and apply it to work activities. The person who is competent in this CA shall be able to adapt to HSE by executing his/her duties while observing Health, Safety and Environment Rules and Regulations.

PENERANGAN/INFORMATION:

1. Safety And Health Hazard Assessment

"Hazard assessment" is the process (required by law) of identifying the hazards associated with defined task, prescribing personal protective equipment and other

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relevant protection measures which must be employed to reduce the risk from the hazards.

1.1 The Hazard Identification And Assessment Methodology

The hazard identification and assessment methodology shall include -

i. Steps and time frame for identifying and assessing the hazards.

One must define the steps for the identification of hazards and a time frame for this identification. The following information should be included -

- a. who will be responsible for the identification: for example, it may be the work place health and safety committee, or an individual or individuals appointed by the committee;
- b. the way in which the identification reports are processed: for example, they may be compiled and processed by the committee, or by individuals appointed by the committee; and
- c. The identification time frame: for example, the identification of hazards for workshop A must be completed in December, for workshop B in April and for workshop C in November.

ii. The keeping of a record of the hazards.

After having identified the hazards, one must establish and maintain an identification record, either in print or electronic format.

iii. A time frame for reviewing and, if necessary, revising the methodology.

The date for the review of the identification: for example, the review of the identification method will be carried out every three years. To complete hazard identification, one can use techniques to identify hazards. Some examples of techniques include, but are not limited to -

- a. work place inspections;
- b. task safety analysis or job hazard analysis;
- c. preliminary investigations;
- d. potential accident factors;
- e. failure analysis;

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- f. accident and incident investigations.

It is in your interest to adopt your own process and your own identification techniques so that they match owner management procedures and the size of business. In fact, the identification method may vary depending on the size of the work place.

1.2 Analyze And Estimate Risk

Using the list of hazards the team undertakes the risk analysis by identifying possible consequences and the likelihood of them occurring. This will help to assess the likelihood of an event occurring and the potential exposure. There are many ways an exposure or event can be initiated. It is important that all causes are considered.

In the analysis, include all who could be affected, such as contractors, or visitors. Highlight those groups or individuals particularly at risk, and take into account and objectively assess the effectiveness of existing control measures.

To analyse risk the team needs to consider the chance of the hazardous situation occurring - the likelihood - and the extent of the harm that would result - the consequence.

For each hazard, the team needs to determine:

- i. How people are exposed;
- ii. Likelihood of exposure happening;
- iii. Frequency of exposure (when and how often), e.g. intermittent (decanting a hazardous substance), or continuous (noise); and
- iv. Consequences of exposure, e.g. fatality, serious injury or permanent health impairment, machinery damage, minor injury or reversible health effect, lost production time.

1.3 Risk Assessment

Risk assessment involves comparing the level of risk identified during the risk analysis process with previously established risk criteria, and ranking the risks in

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order of severity. What the team is trying to do is assess all of the risks that each hazard presents so it can be controlled. Once the risks have been assessed and ranked, the action to eliminate or reduce the risk should be identified and prioritised according to the assigned risk value.

2. Permit To Work Procedure

Permits to work are an important means of fulfilling the Company's general duty to ensure the health and safety of employees. Non-routine work, such as maintenance, cleaning, equipment installation and refurbishment, can produce health and safety risks over and above those normally encountered in the workplace. To control these risks "permits-to-work" are being introduced for the following work activities:

- i. hot work, that is any hot work outside of the controlled workshop environment which could reasonably be foreseen as posing a hazard to others or plant and equipment.
- ii. machinery permit, required where dangerous parts of plant or machinery could reasonably be foreseen as a hazard
- iii. confined spaces, entry into any area where by reason of its enclosed nature there arises a foreseeable risk from the following hazards;
 - a. Injury to any person from fire or explosion
 - b. Loss of consciousness arising from raising the body temperature.
 - c. Loss of consciousness or asphyxiation of any person arising from gas, fume, vapour or the lack of oxygen.
 - d. Downing caused by an increase in the level of a liquid.
 - e. Asphyxiation in a free flowing solid or the inability to reach a respirable environment due to entrapment by a free flowing solid.

A confined space permit must also address the specific requirements in the Industrial Code of Practice for Safe Working in a Confined Space 2010 (ICOP 2010)

- iv. electrical work, all work on installations which poses a hazard from electrocution, fire and explosion

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- v. excavation work, any excavation work, other than the winning of minerals, this permit is required to ensure no underground services pose a risk to those working and to preserve the structural integrity of nearby structures.
- vi. equipment disjoining, used to control the disconnection of any equipment which has contained a liquid or gas i.e. pipe work containing heated bitumen, large diesel tanks etc.
- vii. work at height / roof access, access to fragile roofs or where the roof has no safety edge protection or parapet, any work involving the risk of falling more than 2m.

It is the intention of the company to introduce new or modify existing permits to work as necessary and to review their use as appropriate. The permit to work system applies to all the above work conducted by the Company, it's employees, contractors and all visitors are expected to comply with the requirements of any permits that are in force.

For all operation involving flame, welding and hot cutting This permit is valid only for the job described and the timescales provided		
Description of work		
Location of Work		

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Building		Floor		Room		Location	
Date required (max duration 1 day)			Valid From (time)			To	
Contact Details (method of Contact)							
Mobile Phone		Site telephone		Co. Office No.			
Estates Project Officer		Estates Help Desk		024765 22567		Security	02476522083
Potential Hazards		☐ Oxygen enrichment ☐ Oxygen depletion ☐ Toxic Gas ☐ Explosive Gas ☐ Bio Hazard ☐ Poor lighting ☐ Heat ☐ Noise ☐ Tripping/falling/striking objects					
		☐ Other- provide details					
Control Measures		☐ Hazard / equipment Isolated ☐ Protective equipment required - specify:				☐ Department staff informed	
Other Identified Hazards				Controls Measures			
Mandatory Safety Requirements (<i>See reverse for further guidance</i>)							Actioned
All areas to be checked and combustibles removed or protected before commencement of work							
All areas to be screened, protected, roped off as necessary and warnings signs displayed							
All systems associated with the work to be isolated, inclusive of smoke alarms							
Assistant to standby with fire extinguisher suitable for task. (Competent in use)							
Building Facilities Manager notified							
Area to be checked/inspected for combustion 1 Hour after completion of work							
Person entering work area							
Permit issued by		Date		Time			
Permit Received by		Date					
Permit cancellation (Estates Dept.)							
Name		Date		Time			

Figure 1: Example of Work Permit for Hot Work

3. Method Of Practicing Work Instruction, Safe Operating Procedure And Emergency Response Plan

Safe working instructions outline the recommended safe method of undertaking the process or activity. Written safe working instructions are an essential part of a safe

system of work and are an important part of an overall occupational health and safety program. Safe work instructions do not replace the requirement for training; they may however be used to supplement or guide the training process and provide reminders for staff and students on the operating protocols and controls. The developments of safe working instructions are a local management responsibility. They should be an integral part of the organizational area's induction and refresher training programs.

3.1 Difference between Work Instructions and Safe Operating Procedures

Work Instructions tell how to do something. They are very specific instructions, for example, how to complete a form. It is a tool provided to help someone to do a job correctly. This simple statement implies that the purpose of the work instruction is quality and that the target user is the worker.

Safe Operating Procedures tell you who does what when. In other words, it identifies who completes the form and when this should occur. Safe operating procedures are written guidelines for procedures and tasks involving recognized hazards. They provide the tools for teaching how to work consistently with a maximum degree of efficiency and safety.

In simple terms:

Procedures tell you Who, What and When Work Instructions tell you How
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3.2 Prepare An Emergency Response Plan

Emergencies and disasters can occur any time without warning. The more you are prepared for them, the better you will be able to act, minimizing panic and confusion when an emergency occurs. Training staff who are involved in emergency planning and response is fundamental to an organisation's ability to handle any type of emergency.

Planning for emergencies cannot be considered reliable until it is exercised and has proved to be workable, especially since false confidence may be placed in the integrity of a written plan. Generally, participants in exercises should have an awareness of their roles and be reasonably comfortable with them, before they are subject to the stresses of an exercise. Exercising is not to catch people out. It

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tests procedures, not people. If staffs are under-prepared, they may blame the plan, when they should blame their lack of preparation and training. An important aim of an exercise should be to make people feel more comfortable in their roles and to build morale.

TITLE of ELEMENT		
WORK INSTRUCTION 01-01-01	Revision:	Page

1.0	PURPOSE
1.1	(What is the reason for writing this work instruction) Start with the following statement: "The purpose of this document is to provide instructions for
1.2	Be as specific as you need to be to define the purpose of the document.
2.0	SCOPE
2.1	(Define the area or process that this instruction applies to)
3.0	RECORDS
3.1	(Indicate specific type of records to be maintained and where they will be maintained)
4.0	ASSOCIATED DOCUMENTS
4.1	(Reference upward to the procedures that the work instruction is controlled by and reference forms by control number used to create records. Also, reference other documents, standards or information that are relevant.)
5.0	DEFINITIONS
5.1	(Define any terms and phrases that might be unique to this instruction or to this particular industry)
6.0	RESPONSIBILITY
6.1	(List department(s) or functions responsible for maintaining these instructions)
7.0	TOOLS, GAGES, FIXTURES
7.1	(List all unique tools required to perform tasks as indicated in this work instruction)
8.0	SAFETY REQUIREMENTS
8.1	(List all necessary safety precautions that employees must comply with and all safety equipment that must be used while following the operations specified in this work instruction)
9.0	INSTRUCTIONS
9.1	(List all steps to be taken to complete the scope of this work instruction, in the exact sequence necessary. Work instructions may be in the form of flow charts, bullet instructions, text, photos, digitized images, numbered instructions or any combination of all, as long as the instructions are clear, crisp and do the job as intended)

Figure 2: Example of Work Instruction

SAFETY OPERATING PROCEDURES

'Job Title'

DO NOT use this machine unless a teacher has instructed you in its safe use and operation and has given permission.

**PRE-OPERATIONAL SAFETY CHECKS**

1. Check unit for loose/missing nuts, bolts and screws. Tighten and/or replace as needed.
2. Inspect fuel lines, tank, and area around carburetor for fuel leaks. Do not operate machine if leaks are found.
3. Do not use any attachment or accessory unless it is recommended in the Operator's Manual.
4. Ensure that all guards are fitted, secure and functional.
5. Faulty equipment must not be used. Report suspect machinery immediately

OPERATIONAL SAFETY CHECKS

1. Ensure that no person or animal is endangered when operating equipment.
2. When starting stand the machine upright on a level surface. Check that blower pipe is not blocked by the ground or by any objects.
3. Allow the machine to warm up at fast idle for a few minutes before using.
4. Keep a firm grip. Hold handle with fingers together encircling handle.
5. Maintain a straight wrist position. Avoid using your wrist in a bent, extended or twisted position.
6. The muffler side of the engine should be away from your body to avoid burns.

HOUSEKEEPING

1. Remove any foreign material from engine, air intake and guards.
2. Keep the work area and implement shed in a clean and tidy condition.

POTENTIAL HAZARDS

Figure 3: Example of Safety Operating Procedure

3.3 Types Of Exercises

There are 3 main types of exercise:

a. Discussion-based exercises

Discussion-based exercises are cheapest to run and easiest to prepare. They can be used at the policy formulation stage as a 'talk-through' of how to finalise the plan. More often, they are based on a completed plan and are used to develop awareness about the plan through discussion. In this respect, they are often used for training purposes.

b. Table top exercises

Table top exercises are based on simulation, not necessarily literally around a table top. Usually, they involve a realistic scenario and a time line, which may be real time or may speed time up. Usually table tops are run in a single room, or in a series of linked rooms which simulate the divisions between responders who need to communicate and be co-ordinated. The players are expected to know the plan and they are invited to test how the plan works as the scenario unfolds. This type of exercise is particularly useful for validation purposes, particularly for exploring weaknesses in procedures. Table-top exercises are relatively cheap to run, except in the use of staff time. They demand careful preparation.

c. Live exercises

Live exercises are a live rehearsal for implementing a plan. Such exercises are particularly useful for testing logistics, communications and physical capabilities. They also make excellent training events from the point of view of experiential learning, helping participants develop confidence in their skills and providing experience of what it would be like to use the plan's procedures in a real event. Where the latter purposes are, in fact, the main objective of the exercise, then it is essentially a training exercise or practice

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drill. Live exercises are expensive to set up on the day and demand the most extensive preparation.

4. Specific Personal Protective Equipment (PPE)

PPE is defined as all equipment (including clothing affording protection against the weather) which is intended to be worn or held by a person at work and which protects him against one or more risks to his health or safety', e.g. safety helmets, gloves, eye protection, high-visibility clothing, safety footwear and safety harnesses.

4.1 Assessing Suitable PPE

To allow the right type of PPE to be chosen, carefully consider the different hazards in the workplace. This will enable you to assess which types of PPE are suitable to protect against the hazard and for the job to be done. Ask your supplier for advice on the different types of PPE available and how suitable they are for different tasks. It may be necessary in a few particularly difficult cases to obtain advice from specialist sources and from the PPE manufacturer.

Consider the following when assessing whether PPE is suitable:

- i. Is it appropriate for the risks involved and the conditions at the place where exposure to the risk may occur? For example, eye protection designed for providing protection against agricultural pesticides will not offer adequate face protection for someone using an angle grinder to cut steel or stone.
- ii. Does it prevent or adequately control the risks involved without increasing the overall level of risk?
- iii. Can it be adjusted to fit the wearer correctly?
- iv. Has the state of health of those who will be wearing it been taken into account?
- v. What are the needs of the job and the demands it places on the wearer? For example, the length of time the PPE needs to be worn, the

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physical effort required to do the job and the requirements for visibility and communication.

- vi. If more than one item of PPE is being worn, are they compatible? For example, does a particular type of respirator make it difficult to get eye protection to fit properly?

4.2 The Hazards And Types Of PPE

- i. Eyes

Hazards: chemical or metal splash, dust, projectiles, gas and vapour, radiation. Options: safety spectacles, goggles, face shields, visors.

- ii. Head

Hazards: impact from falling or flying objects, risk of head bumping, hair entanglement. Options: a range of helmets and bump caps.

- iii. Breathing

Hazards: dust, vapour, gas, oxygen-deficient atmospheres.

Options: disposable filtering face piece or respirator, half- or full-face respirators, air-fed helmets, breathing apparatus.

- iv. Protecting the body

Hazards: temperature extremes, adverse weather, chemical or metal splash, spray from pressure leaks or spray guns, impact or penetration, contaminated dust, excessive wear or entanglement of own clothing. Options: conventional or disposable overalls, boiler suits, specialist protective clothing, e.g. chain-mail aprons, high-visibility clothing.

- v. Hands and arms

Hazards: abrasion, temperature extremes, cuts and punctures, impact, chemicals, electric shock, skin infection, disease or contamination.

options: gloves, gauntlets, mitts, wrist cuffs, armlets.

- vi. Feet and legs

Hazards: wet, electrostatic build-up, slipping, cuts and punctures, falling objects, metal and chemical splash, abrasion.

Options: safety boots and shoes with protective toe caps and penetration-resistant mid-sole, gaiters, leggings, spats.

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4.3 Training

- i. Make sure anyone using PPE is aware of why it is needed, when it is to be used, repaired or replaced and its limitations.
- ii. Train and instruct people how to use it properly and make sure they are doing this.
- iii. Because PPE is the last risk control option after other methods of protection have been considered, it is important that users wear it all the time they are exposed to the risk. Never allow exemptions for those jobs which take 'just a few minutes'.
- iv. Check regularly that PPE is being used and investigate fully any reasons why it is not. Safety signs can be useful reminders to wear PPE.

4.4 Maintenance

Make sure equipment is:

- i. well looked after and properly stored when it is not being used, for example in a dry, clean cupboard, or in the case of smaller items, such as eye protection, in a box or case;
- ii. kept clean and in good repair - follow the manufacturer's maintenance schedule (including recommended replacement periods and shelf lives). Simple maintenance can be carried out by the trained wearer, but more intricate repairs should only be done by specialists.

Make sure suitable replacement PPE is always readily available.

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SOALAN/QUESTION :

1. What is done at the risk assessment ?

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2. List five work activities that require a work permit

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3. Why training the use of PPE is important ?

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RUJUKAN / REFERENCES:

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